Kaustubh Dwivedi

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OBJECTIVE

Software engineering student with production experience at IIT Madras. Skilled in Python, JavaScript, Flask, SQL, REST APIs, and modern frontend development, seeking Software Development Engineer or Full Stack roles.

EDUCATION

Indian Institute of Technology Madras — **Chennai, Tamil Nadu**
BS- Data Science — *Graduation Date: May 2027*
Percentage: 80.01%

Kendriya Vidyalaya — **Lucknow, Uttar Pradesh**
Class 12-Science — *2022*
Percentage: 81.50%

PROJECT

Vehicle Parking Web Application | Github | Live | Demo

- Built a full-stack parking management system with role-based access control (RBAC), real-time parking spot availability, vehicle reservations, and automated billing based on parking duration.
- Designed and implemented RESTful APIs using Flask, integrated secure authentication, asynchronous background jobs, and generated monthly activity and usage reports.

Quiz Master Application | Github | Live | Demo

- Developed a role-based quiz management system using Python and Flask, supporting admin and student roles, subject and chapter management, quiz scheduling, timed assessments, score tracking, and performance analytics.
- Implemented secure user authentication, authorization, CSRF protection, and encrypted password storage to ensure application security and data integrity.

AI Data Analyst Agent | Live | Github

- Built a natural-language data analyst where the LLM chooses from a fixed set of pandas tools and pandas computes every number, preventing hallucinated figures; supports follow-up questions and charts on request.
- Designed a bounded, self-correcting agent that flags any answer figure not found in the computed data; deployed as Dockerized FastAPI and Streamlit services on Render with 3 LLM providers and 412 tests.

EXPERIENCE

Software Engineer Intern
Sudha Gopalakrishnan Brain Centre (SGBC), IIT Madras | May 2026 – Sep 2026

- Built and integrated full-stack plugins and features for a production neuroscience research platform using Python, Flask, JavaScript, Vue.js, SQL, REST APIs, and OpenLayers, supporting brain-section visualization, detection analysis, heatmaps, and interactive dataset exploration.
- Developed backend services and API-driven workflows for image registration, ML-based region classification, detection processing, and dynamic job loading; integrated asynchronous processing with Celery/Redis and database-backed application workflows.
- Resolved complex production issues across frontend, backend, and visualization layers, including event/state synchronization, duplicate component mounting, image-coordinate transformations, IIIF image rendering, heatmap alignment, responsive UI behavior, and plugin interoperability.

SKILLS & INTERESTS

Technical: Python | JavaScript | SQL | Flask | REST APIs | SQLAlchemy | PostgreSQL | Celery | Redis | Vue.js | HTML/CSS | Git | GitHub | Docker | Machine Learning | NumPy

Interests: Badminton| Coding | Playing Cricket| Gaming